BiteFixes Enterprise Architecture Manual

Volume 05

Business Rules Catalog (BRC)

Document ID: BFA-005

Version: 1.0

Status: Draft

Language: English

Architecture Level: Enterprise

Copyright

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Chapter 1

Introduction

This document defines every official business rule used by the BiteFixes Enterprise Platform.

Business rules are **not implemented here**.

This document only specifies them.

Implementation comes later.

Chapter 2

Rule Classification

Every rule has the following structure.

```
Rule ID

Category

Domain

Description

Condition

Action

Severity

Configurable
```

```
Version
```

Example

```
BR-0001

Category

Validation

Domain

Customer

Description

Customer email must be unique.

Severity

Error

Configurable

No
```

Rule Categories

Rules are classified into:

```
Validation

Security

Workflow
```

`Financial`

`Inventory`

`Notification`

`AI`

`Compliance`

`Permission`

`Automation`

Chapter 3

Company Rules

BR-0001

Company must have one default language.

Severity

Critical

BR-0002

Company timezone is mandatory.

BR-0003

Company cannot be deleted.

Status changes to Archived.

BR-0004

Every company owns its own data.

No cross-company access.

BR-0005

Subscription controls available features.

Chapter 4

Customer Rules

BR-1001

Phone number must be unique inside the same company.

BR-1002

Email uniqueness is optional.

Company configuration defines this.

BR-1003

Blocked customers cannot create new tickets.

BR-1004

Inactive customers cannot receive notifications.

BR-1005

Preferred language defaults to company language.

BR-1006

Customer score is calculated.

Never manually edited.

BR-1007

Customer history cannot be deleted.

Only archived.

Chapter 5

Device Rules

BR-2001

Every device belongs to exactly one customer.

BR-2002

Device serial number must be unique.

BR-2003

IMEI uniqueness is enforced when available.

BR-2004

Warranty expiration cannot precede purchase date.

BR-2005

Diagnosis is mandatory before repair.

BR-2006

Repair history is immutable.

BR-2007

Archived devices cannot receive new tickets.

Chapter 6

Service Rules

BR-3001

Every service belongs to one category.

BR-3002

Inactive services cannot be selected.

BR-3003

Estimated repair time must be greater than zero.

BR-3004

Base price cannot be negative.

BR-3005

Warranty days must be defined.

Chapter 7

Ticket Rules

This domain contains the highest number of business rules.

BR-4001

Customer is mandatory.

BR-4002

Device is mandatory.

BR-4003

Service is mandatory.

BR-4004

Ticket must have one status.

BR-4005

Status transitions follow the official workflow.

```
Draft

↓

Open

↓

Diagnosis

↓

Waiting Approval

↓

Waiting Parts

↓
```

```
Repair

↓

Testing

↓

Ready

↓

Delivered

↓

Closed
```

No other transitions are allowed unless defined by policy.

BR-4006

Closed tickets cannot be modified.

BR-4007

Cancelled tickets cannot be reopened.

BR-4008

Warranty is generated only after successful delivery.

BR-4009

Final cost cannot be lower than zero.

BR-4010

Delivery date cannot precede repair completion.

Chapter 8

Knowledge Rules

BR-5001

Knowledge articles require approval.

BR-5002

Only published articles are searchable.

BR-5003

Archived articles remain available for audit.

BR-5004

AI learns only from approved knowledge.

BR-5005

Knowledge always belongs to one service.

Chapter 9

AI Rules

BR-6001

AI never changes business data directly.

BR-6002

AI cannot bypass security policies.

BR-6003

AI recommendations require confidence score.

BR-6004

Low-confidence responses must request clarification.

BR-6005

AI decisions must be auditable.

BR-6006

Every AI response stores:

```
Intent

Confidence

Knowledge Source

Processing Time

Language

Model

Version
```

Chapter 10

Security Rules

BR-7001

Authentication required.

BR-7002

Authorization required.

BR-7003

Permission validation required.

BR-7004

Company isolation mandatory.

BR-7005

Audit logging mandatory.

Chapter 11

Notification Rules

Notifications are event-driven.

Examples:

```
Ticket Created

↓

Customer Confirmation

↓

Technician Notification

↓

Manager Dashboard
```

No notification is sent without an originating event.

Chapter 12

Rule Priorities

Every rule has a priority level.

Level	Meaning
Critical	Must never be violated
High	Required for business integrity
Medium	Operational recommendation
Low	Informational

Chapter 13

Rule Lifecycle

Rules pass through:

```
Draft

↓

Review

↓

Approved

↓

Published

↓

Deprecated

↓

Archived
```

No rule is deleted.

Chapter 14

Rule Versioning

Each rule contains:

`Rule ID`

`Version`

`Effective Date`

`Author`

`Reviewer`

`Approval Date`

This allows historical tracking.

Chapter 15

Rule Engine Vision

The ultimate goal is to externalize business rules into a configurable engine.

Example:

`IF`

`Customer = VIP`

`AND`

```
Repair Cost > 1000

THEN

Manager Approval Required
```

The backend interprets these rules instead of hardcoding them.

Chapter 16

Rule Governance

A Business Rule Board (or architecture team) should approve any change to published rules.

Every modification must answer:

- Why is the rule changing?
- Which domains are affected?
- Is it backward compatible?
- Does it impact existing workflows?
- Does it require database changes?

Chapter 17

Rule Traceability

Every business rule must be traceable to:

- Business requirement.
- Domain.
- Use case.

- API.
- Test case.
- Documentation.
- Audit record.

This ensures complete alignment across the platform.

Conclusion

The Business Rules Catalog is the functional constitution of BiteFixes.

The application code is merely an implementation of these rules.

Any conflict between code and the BRC must be resolved in favor of the BRC.

Architect Recommendation

A partir de este punto, comenzaría a diseñar el **Workflow Engine Specification (WFES)**.

Hasta ahora sabemos **qué reglas existen**.

El siguiente paso es definir **cómo fluyen los procesos completos**.

Ese documento especificará:

- Máquinas de estado.
- Transiciones.
- Acciones automáticas.
- Temporizadores.
- Escalaciones.
- SLA.
- Reglas condicionales.
- Procesos paralelos.
- Reintentos.

- Compensaciones.
- Integración con Bitey.